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TWAP Algorithm

Function

Breaks down a large order into smaller equal-sized pieces and executes them at fixed time intervals to minimize market impact and achieve a time-weighted average execution price.

Detailed Explanation

The Time-Weighted Average Price (TWAP) algorithm divides the total order Q_{total} over a specified execution time T_{total} into equal parts, executing each segment $Q_{segment}$ at regular intervals $T_{segment}$.

1. Segment Duration ($T_{segment}$):

Divides the total time into $N_{intervals}$ equal parts:

$$T_{segment} = \frac{T_{total}}{N_{intervals}}$$

2. Segment Quantity ($Q_{segment}$):

Splits the total quantity evenly among the intervals:

$$Q_{segment} = \frac{Q_{total}}{N_{intervals}}$$

3. Execution:

At each time t_i , where $t_i = i \cdot T_{segment}$ for $i = 1, 2, \dots, N_{intervals}$, an order of $Q_{segment}$ is placed.

The algorithm ensures smooth execution by spreading orders evenly, reducing price slippage and market disruption.

Real-Life Example

Scenario: An institutional investor wants to buy **1,200,000 shares** of ABC stock over **6 hours**, divided into **12 intervals**.

Inputs

- $Q_{total} = 1,200,000$
- $T_{total} = 6 \text{ hours}$
- $N_{intervals} = 12$

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- Segment quantity: 10 segments 12 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Execution

Every 30 minutes, a market order for **100,000 shares** is placed. This continues until all 12 intervals are executed, minimizing the market impact and achieving a smooth execution process.

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